

Find the value of each of the following limits (if they exist).

1. $\lim_{x \rightarrow 3} \frac{x^3 - 27}{x - 3}$

2. $\lim_{x \rightarrow 1} \frac{3x - \sqrt{2x + 7}}{x - 1}$

3. $\lim_{x \rightarrow 3^-} \frac{|3x - 9|}{3 - x}$

4. $\lim_{x \rightarrow 2} f(x)$ if $f(x) = \begin{cases} 2x + 3 & \text{for } x \in (-\infty, 2] \\ 3x - 1 & \text{for } x \in (2, \infty) \end{cases}$

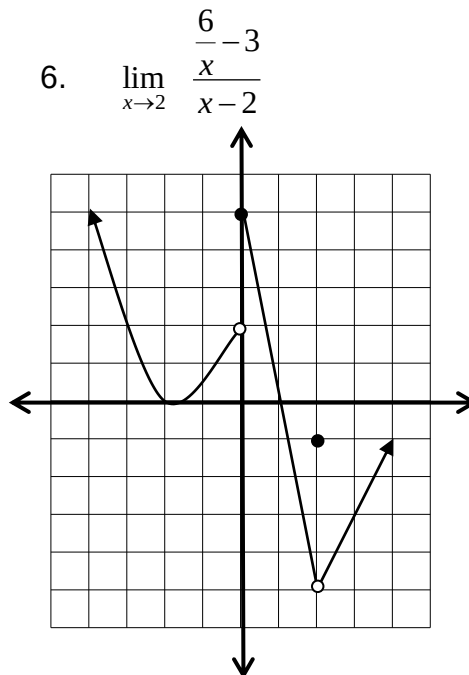
5. $\lim_{x \rightarrow 1} \frac{2x^2 + 3x - 5}{3x^2 - 4x + 1}$

6. $\lim_{x \rightarrow 2} \frac{\frac{6}{x} - 3}{x - 2}$

7. a) $\lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}}$

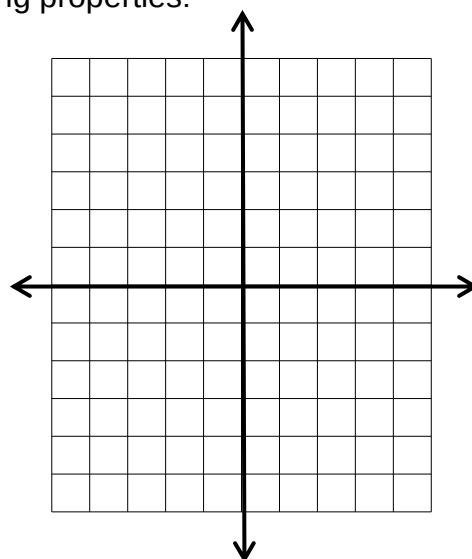
b) $\lim_{x \rightarrow 0^+} f(x) = \underline{\hspace{2cm}}$

c) $\lim_{x \rightarrow -3} f(x) = \underline{\hspace{2cm}}$



8. Sketch a graph of a function with all of the following properties:

- $f(-2) = 3$
- $\lim_{x \rightarrow 1^+} f(x) = 3$
- $\lim_{x \rightarrow 1^-} f(x) = -1$
- $f(2) = 2$
- $\lim_{x \rightarrow \infty} f(x) = -4$
- $\lim_{x \rightarrow -\infty} f(x) = -\infty$
- The graph is continuous over the intervals $(-\infty, 1)$ and $(1, \infty)$



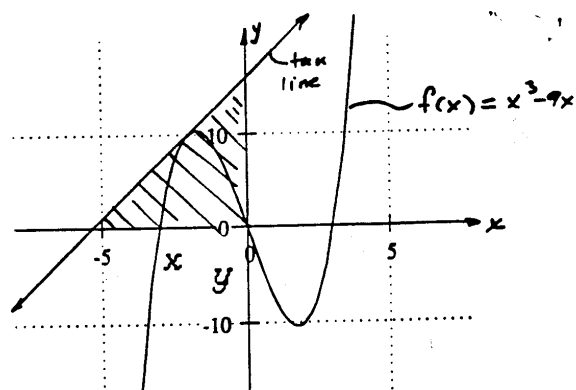
9. Find $\lim_{x \rightarrow 27} \frac{x - 27}{\sqrt[3]{x} - 3}$

10. Find $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^3 - x^2 - 14x + 24}$

11. Find the equation of the tangent line to the function $f(x) = \frac{6}{x-3}$ at $(1, -3)$.
12. A construction worker loses his shoe while working on the roof of a high rise building. After t seconds, the shoe has fallen a distance of s meters, where $s(t) = 320 - 5t^2$, $0 \leq t \leq 8$.
- a) Find the average velocity of the shoe during the first 3 seconds of the fall.
 - b) Find the velocity of the shoe at time $t = 5$ seconds.
13. Explain why $\lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.
14. Can $\lim_{x \rightarrow 2} f(x) = 3$ if $f(2) = 4$? Explain.
15. If $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{g(x)}{x} = 2$, evaluate:
- a) $\lim_{x \rightarrow 0} g(x)$
 - b) $\lim_{x \rightarrow 0} \frac{g(x)}{f(x)}$

16. The graph of $f(x) = x^3 - 9x$ is as shown. A tangent line has been drawn at the point $(-2, f(-2))$. You'll notice that the tangent line intersects both the x axis and the y axis. Find the area of the triangle formed in the 2nd quadrant by the tangent line and the axes. (The area of a triangle is given by

$$A = \frac{1}{2}(\text{base})(\text{height}).$$



17. Find the derivative of $f(x) = \frac{\sqrt{x+2}}{x^2}$.

18. Find the derivative of each of the following functions.

a) $a(x) = 7x^3 + \frac{3}{2}x^2 + 3x - 8$

b) $b(x) = \frac{2}{x^2} + \frac{2}{x} + 3 + 5x + 2x^2$

c) $c(x) = (x^2 + 3x + 2)^4$

d) $d(x) = \frac{3x^2 + 1}{4 - 3x}$

e) $e(x) = \frac{5}{\sqrt[3]{2x^4 - 1}}$

f) $f(x) = (x+2)^2(2x-1)^3$

g) $g(x) = (2x-7)\sqrt{3x^2-5x}$

h) $h(x) = \sqrt{\frac{1+x}{1+x^2}}$

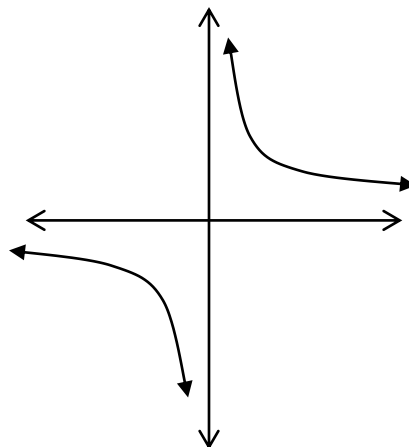
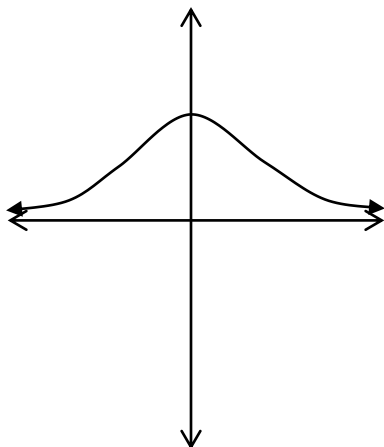
19. At what point(s) is the slope of $y = x^3 - 3x^2 - 4x + 7$ equal to 5?

20. If $f'(a) = \lim_{h \rightarrow 0} \frac{\sqrt{4+h} + \frac{5}{(4+h)^2} - \frac{37}{16}}{h}$, find $f(x)$ and a .

21. Let $v = h(p(u))$ where $p(u) = u^{\frac{3}{2}} + 3u^3 - 5u$. If $h'(-1) = 5$, find $\frac{dv}{du}$ when $u = 1$.

22. **OMIT** Find the equation of the tangent to the ellipse $x^2 + xy + y^2 = 13$ at $(3,1)$. **OMIT**

23. Show a sketch of the derivative function for each of the following:



24. Given that $y = \frac{x}{1+x^2}$, show that $\frac{d^2y}{dx^2} = \frac{2x^3 - 6x}{(1+x^2)^3}$

25. Let $f(x) = \left\{ \left[(1-x)^4 + x^4 \right]^4 + x^4 \right\}$. Find $f'(0)$ and $f'(1)$

26. The number N of new *SNAZ* cars sold per week in Canada, t weeks after going on the market, is $N(t) = \frac{300t^2}{1+t^2}$, $0 \leq t \leq 10$

- At what rate is the number of sales changing after one week? After 5 weeks?
- Does the number of sales per week decrease at any time during this 10 week period? Explain.

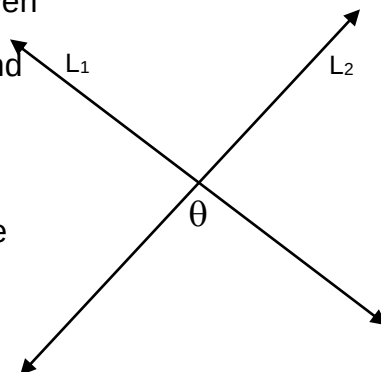
27. The tangent of the angle θ formed when 2 non-vertical lines L_1 and L_2 intersect is given

by the formula $\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$ where m_1 and

m_2 are the slopes of the lines L_1 and L_2

respectively. If $f(x) = x^3 + x$ and

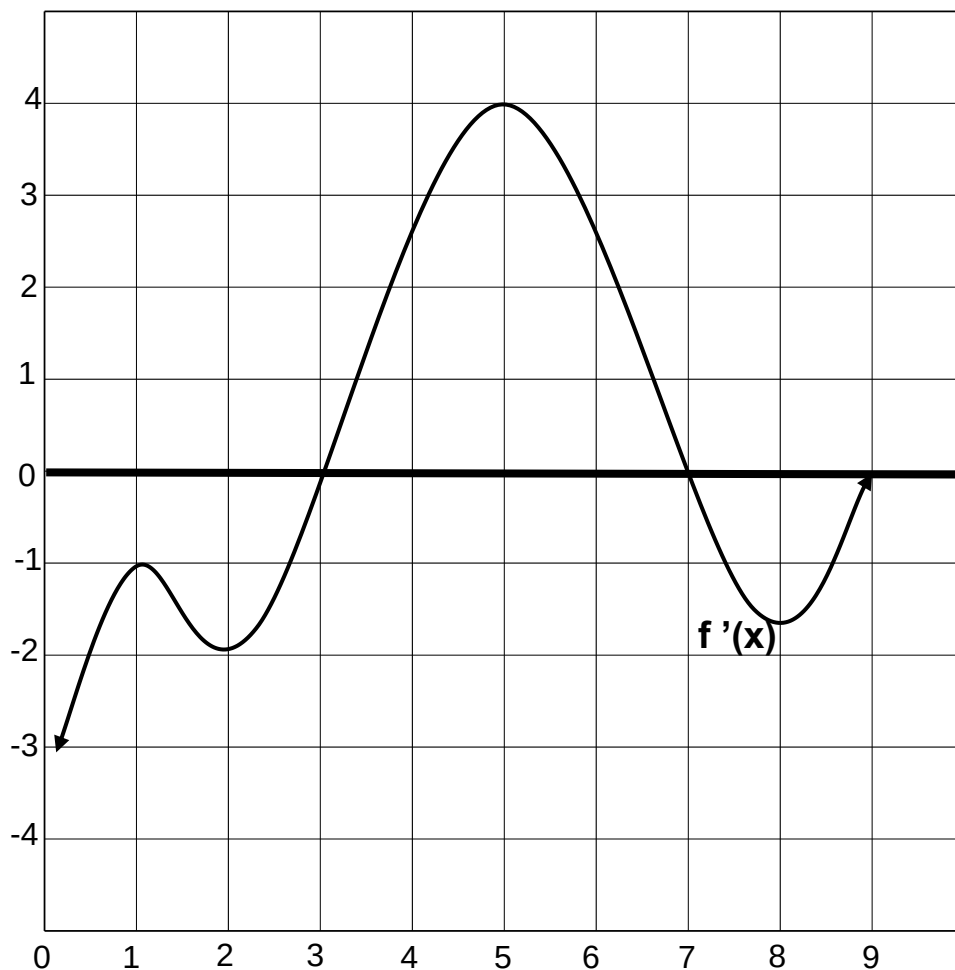
$g(x) = x^2 + 1$, what is the angle between the tangents to these graphs at the point of intersection?



28. a) $\lim_{x \rightarrow 2^-} \frac{x}{x-2}$ b) $\lim_{x \rightarrow \infty} \frac{x^2-1}{(3x+2)(2x-5)}$

c) $\lim_{x \rightarrow \infty} \left(\sqrt{9x^2 + 5x - 4} - 3x \right)$

29. The graph of the derivative of a function $f(x)$ is shown below. Use the graph of $f'(x)$ to answer the following questions about $f(x)$.



- a) On what interval(s) is $f(x)$ increasing?
- b) On what interval(s) is $f(x)$ concave down?
- c) Which values of x correspond to inflection points on $f(x)$?
- d) Which values of x correspond to local maximum point(s) on $f(x)$?

30. Sketch the graph of a continuous function that satisfies all these conditions.

- $f(0) = f(3) = 2, \quad f(-1) = f(1) = 0$
- $f'(-1) = f'(1) = 0$
- $f'(x) < 0$ for $x \in (-\infty, -1)$ and for $x \in (0, 1)$
- $f'(x) > 0$ for $x \in (-1, 0)$ and for $x \in (1, \infty)$
- $f''(x) > 0$ for $x < 3$ ($x \neq 0$), and $f''(x) < 0$ for $x \in (3, \infty)$
- $\lim_{x \rightarrow \infty} f(x) = 4, \quad \lim_{x \rightarrow -\infty} f(x) = \infty$

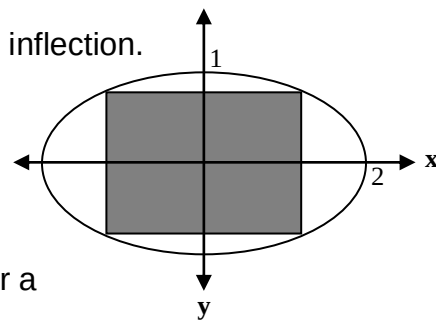
31. For a certain function it is known that $f''(x) = \frac{2x(x-5)}{(x+3)^3}$ and that there is a vertical asymptote at $x = -3$.

- a) Determine the intervals where the graph is concave up and where it is concave down.
- b) Determine the x coordinates of any inflection points.

32. Sketch the graph of $f(x) = x^4 - 4x^3$. Clearly label all minimum and maximum points as well as all intercepts and points of inflection.

33. Sketch the graph of $f(x) = \frac{4x^2}{x^2 + 3}$. Clearly identify all asymptotes, intercepts, minimum and maximum points, and any points of inflection.

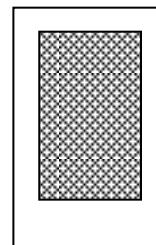
34. Find the dimensions of the largest rectangle with sides parallel to the axes that can be inscribed in the ellipse $x^2 + 4y^2 = 4$ as shown in the figure.



35. Find the value(s) of x which make the function a maximum or a minimum

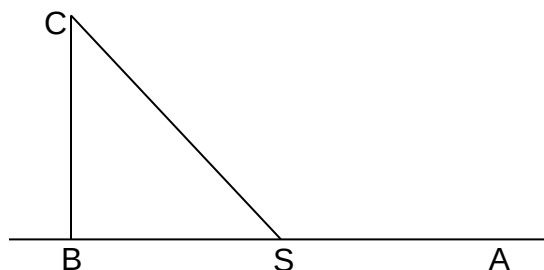
given that $f'(x) = \frac{x}{\sqrt{x^2 + 25}} + \frac{x-6}{\sqrt{x^2 - 12x + 85}}$ and $x \in [0, 6]$.

36. A printed page will have margins of 2 cm at the top and sides, and 4 cm at the bottom. If the printed area is 150 cm^2 , find the dimensions of the whole page so that its area will be a minimum.



37. Find 2 numbers that have a sum of 15 if the product of the cube root of one of them and the square root of the other is to be a maximum.

38. A powerline is to be constructed from a power station at point A to an island at point C which is 1 km out in the water from a point B on the straight shoreline. Point B is 4 km downshore from the power station at A. It costs \$5000 per km to lay the power line under the water and \$3000 per km to lay the line under ground. At



what point S, downshore from A should the line come to the shore in order to minimize the cost? What is that minimal cost?

39. A wire 17 cm long is cut into two pieces. One piece is bent to form a square and the other is bent to form a rectangle that is twice as long as it is wide. How should the wire be cut in order to minimize the total area of the square and rectangle?

40. What are the dimensions of the cone of maximum volume that can be inscribed inside a sphere of radius 10 cm.

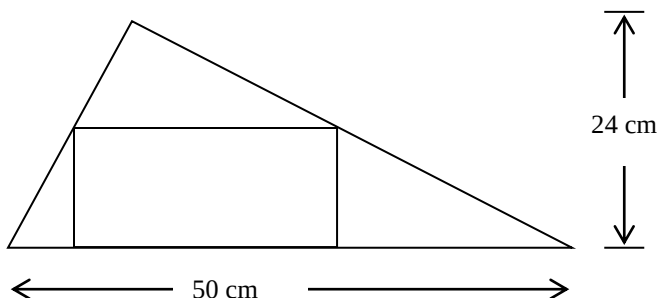
$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h \quad V_{\text{sphere}} = \frac{4}{3}\pi r^3$$

41. Find the dimensions of the circular cylinder of largest volume that can be inscribed in a right circular cone of height 15 inches and radius 6 inches.

42. A hockey team plays in an area with a seating capacity of 15,000 spectators. With ticket prices at \$12, average attendance at a game has been 11,000. A market survey indicates that for each dollar that ticket prices are lowered, the average attendance will increase by 1000. How should the owners of the team set ticket prices to maximize their revenue from ticket sales?

43. A tin can with a closed top is to be constructed to contain 100 cm^3 . Find the radius that will minimize the amount of metal used.

44. Determine the dimensions of the rectangle of maximum area that can be inscribed in the given triangle.



45. Simplify: $\ln \sqrt{\frac{1}{e^{4x}}}$

46. Find x to 3 decimal places if $\ln(\ln 7x) = -1$

47. Find $\frac{dy}{dx}$ for each of the following. Simplify your answers fully.

a) $y = \log_8 e^x$

b) $y = \log(x^2 + 1)$

c) $y = 6 \ln e^{2x} - e^{4 \ln x}$

d) $y = \sqrt{\ln e^x} - \ln e^{\sqrt{x}}$

e) $y = 3e^{5x} + \ln\left(\frac{5}{x^2}\right)$

f) $y = e^{x^2} - (e^x)^2$

g) $y = \ln \sqrt{x} - \sqrt{\ln x}$

h) $xe^y + y - 2x = \ln 2$

i) $y = \frac{e^x}{1 + e^{-3x}}$

j) $y = \left(\frac{x}{2}\right)^{4x}$

48. Find and classify the local maximum/minimum points of the function $f(x) = x(\ln x)^3$.

49. The public works department of a town counted 400 rats one year. It is predicted that the rat population will grow exponentially for the next few years. If t is time, in years, then the growth constant for the rat population is estimated to be 0.018. (Remember that $P(t) = P_0 e^{kt}$). Find the rate of change of the rat population after 10 years.

50. For what values of A does the function $y = e^{Ax}$ satisfy the equation $y'' + 3y' + 2y = 0$?

Answers:

1. 27

2. $\frac{8}{3}$

3. -3

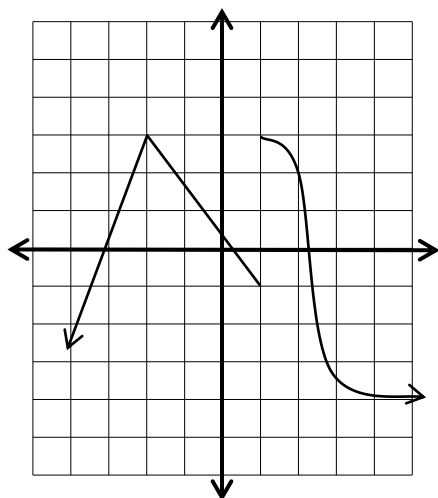
4. DNE

5. $\frac{7}{2}$

6. $\frac{-3}{2}$

7. a) -5 b) 5 c) 2

8.



9. 27

10. $\frac{5}{7}$

11. $3x + 2y + 3 = 0$

12. a) -15 m/s b) -50 m/s

13. The function is not approaching the same value from both sides of 0.

14. Yes. It happens in the case of a hole where the function is redefined.

15. a) 0 b) 2

16. $\frac{128}{3}$

17. $\frac{-3x-8}{2x^3\sqrt{x+2}}$

18. a) $21x^2 + 3x + 3$

b) $\frac{-4}{x^3} - \frac{2}{x^2} + 5 + 4x$

c) $4(2x+3)(x^2+3x+2)^3$

d) $\frac{-9x^2+24x+3}{(4-3x)^2}$

e) $\frac{-40x^3}{3}(2x^4-1)^{\frac{-4}{3}}$

f) $10(x+2)(x+1)(2x-1)^2$

g) $\frac{1}{2}(3x^2-5x)^{\frac{-1}{2}}(24x^2-72x+35)$

h) $\frac{1}{2}\left(\frac{1+x}{1+x^2}\right)^{\frac{-1}{2}}\frac{1-2x-x^2}{(1+x^2)^2}$

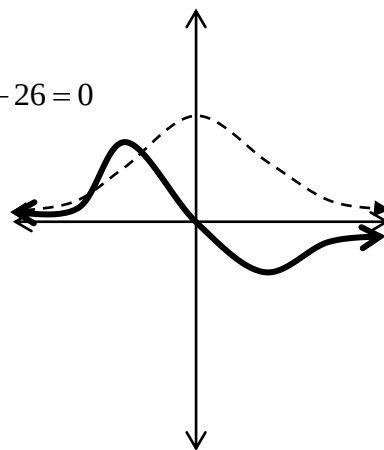
19. (3, -5) and (-1, 7)

20. $f(x) = \sqrt{x} + \frac{5}{x^2} + C$ and $a = 4$

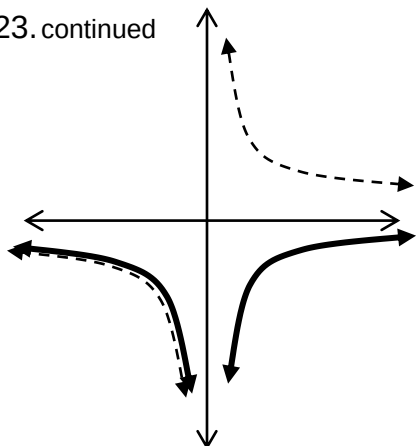
21. $\frac{55}{2}$

22. $7x + 5y - 26 = 0$

23.



23. continued



24. you'll know if you've done it right because the answer is given.

25. $f'(0) = -64$ and $f'(1) = 640$

26. a) 150 cars/week ; 4.4 cars/week

b) No. $N'(t) = \frac{600}{(1+t^2)^2}$ and that is

always positive, the number of cars sold continues to rise.

27. 167.5° or 12.5°

28. a) $-\infty$ b) $\frac{1}{6}$ c) $\frac{5}{6}$

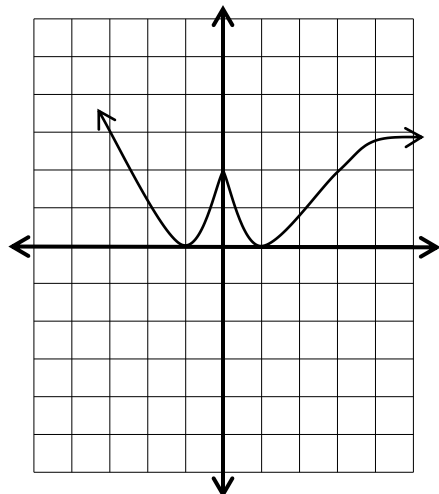
29. a) (3,7)

b) $(1,2) \cup (5,8)$

c) 1, 2, 5, 8

d) 7

30.

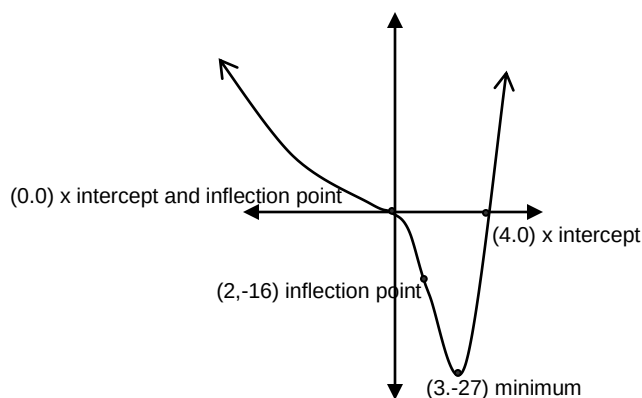


31. a) concave up: $(-3,0) \cup (5,\infty)$

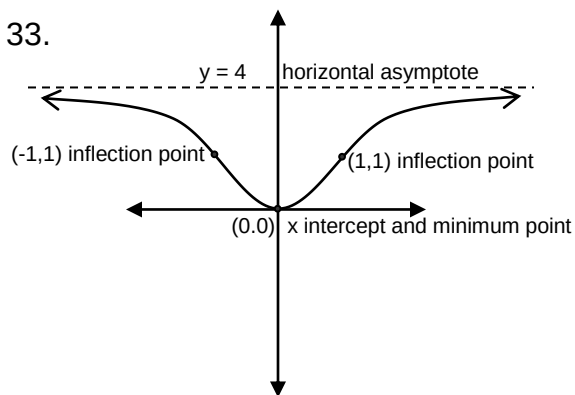
concave down: $(-\infty,-3) \cup (0,5)$

c) 0 and 5

32.



33.



34. $\sqrt{2} \times 2\sqrt{2}$

35. $x = \frac{5}{2}$

36. 14 cm wide by 21 cm tall

37. Take the square root of 9 and the cubed root of 6.

38. S should be $\frac{3}{4}$ km from B and

$\frac{13}{4}$ km from A. The cost will be \$16 000.

39. The maximum area occurs when the rectangle is 3 cm by 1.5 cm and the square is 2 x 2 cm. Cut the wire, therefore, into a piece which is 8 cm long (square) and use the 9 cm piece for the rectangle.

40. radius = $\frac{20\sqrt{2}}{3}$ and height = $\frac{40}{3}$

41. radius = 4 inches and height = 5 inches

42. \$11.50

43. $\sqrt[3]{\frac{50}{\pi}}$ cm

44. 25 x 12 cm

45. $-2x$

46. 0.206

47. a) $\frac{1}{\ln 8}$ or $\log_e 8$ b)

$$\frac{2x}{x^2 + 1} \log e$$

c) $12 - 4x^3$ d) 0 e) $15e^{5x} - \frac{2}{x}$

f) $2xe^{x^2} - 2e^{2x}$ g) $\frac{1}{2x} \left[1 - \frac{1}{\sqrt{\ln x}} \right]$

h) $\frac{2 - e^y}{xe^y + 1}$

i) $\frac{e^x(1 + 4e^{-3x})}{(1 + e^{-3x})^2}$

j) $4 \left(\frac{x}{2} \right)^{4x} \left(1 + \ln \frac{x}{2} \right)$

48. absolute minimum at $\left(\frac{1}{e^3}, \frac{-27}{e^3} \right)$

49. 8.62 rats/year

50. $A = -2$ or $A = -1$